to improve both code maintainability and rendering performance?"

2.3 Leveraging HTML APIs in React development

HTML APIs are collections of protocols and tools that enable developers to create interactive and dynamic web interfaces using HTML elements and attributes. Defined in the HTML specification, these APIs ensure standardization and consistency across web browsers. While HTML APIs are not inherently part of the JavaScript language, they are typically accessed and manipulated through JavaScript, making the two technologies closely interconnected.

Understanding HTML APIs is highly beneficial in practical coding interviews. It demonstrates proficiency with browser fundamentals and the ability to solve problems that go beyond the abstractions provided by frameworks. This expertise allows candidates to implement a wide range of features, such as accessing device hardware (like cameras and geolocation), handling clipboard operations, enabling drag-and-drop functionality, facilitating file access, fetching data from servers, integrating Google Maps or YouTube, leveraging local storage for persistent client-side data, manipulating graphics and multimedia (using Canvas and SVG APIs), managing messaging and notifications, and supporting real-time communication with servers through WebSocket. For a comprehensive list of these APIs and their usage examples, refer to table A.1 in Appendix A.1.

Candidates are not expected to memorize every API, but they should be able to consult the official documentation and use it to complete required tasks. Below are some sample questions illustrating this approach.

Initial question: "How do you use the camera in a web application?"

Analysis: This question assesses a candidate’s ability to practically integrate device hardware—specifically the camera—into web applications. Candidates should demonstrate awareness of secure origins (HTTPS, localhost), cross-browser compatibility, error handling, and clear communication regarding privacy and permissions. The ideal response goes beyond code implementation to address accessibility, user experience, and